The Negative-Viscosity Effect in Ferrofluids

A Complete First-Principles Reconstruction
of Every Equation

VOLUME I

Mathematical and Physical Foundations,
and the Derivation of Equation (SM-1)

A self-contained graduate textbook built on the papers

M. I. Shliomis and K. I. Morozov,
Negative viscosity of ferrofluid under alternating magnetic field,
Physics of Fluids **6**, 2855–2861 (1994)

J.-C. Bacri, R. Perzynski, M. I. Shliomis and G. I. Burde,
"Negative-Viscosity" Effect in a Magnetic Fluid,
Physical Review Letters **75**, 2128–2131 (1995)

Prepared as a continuously growing textbook.
Every subsequent instalment continues this numbering without interruption.

Preface: How This Book Works

What this volume is

This is the first instalment of a single, continuously growing textbook whose final goal is stated once and never modified:

> *Every equation appearing in the two papers listed on the title page will be derived completely, from first principles, with no omitted algebraic step, no appeal to "standard results", and no unexplained symbol.*

Because the physics of the two papers rests on continuum mechanics, statistical mechanics, magnetostatics and rotational Brownian motion — none of which may be assumed known — the book must first build those tools. That is what Volume I does. It assumes only elementary algebra, trigonometry, and the ability to differentiate and integrate a function of one variable. Everything else, including vector calculus, index notation, the Navier–Stokes equations and Stokes' law for a rotating sphere, is constructed here from scratch.

Volume I ends with the first equation of the Shliomis–Morozov paper, Eq. (SM-1), derived in the eight-part format that will be used for every paper equation from here on.

A note on continuity

This document is Volume I: it is the beginning of the derivation project, and no earlier material precedes it. Everything derived here is therefore new, and every later volume will continue from exactly the point at which this one stops, preserving all numbering, notation and conventions established below. The command **Continue** will always produce the next volume, never a restart and never a repetition.

The two numbering systems

To keep cross-references unambiguous forever, two distinct systems are used.

1. **Paper equations** carry the tag of the source paper:
   - **(SM-n)** is equation n of Shliomis & Morozov (1994), $n = 1, \dots, 43$;
   - **(B-n)** is equation n of Bacri, Perzynski, Shliomis & Burde (1995), $n = 1, \dots, 10$.

   These tags never change, in any volume.

2. **Derived and intermediate equations** are numbered **(chapter.section.number)**, e.g. (6.3.11). Every intermediate step of every derivation receives such a number so that it can be referred to later.

The eight-part format for paper equations

Every equation taken from the papers is presented in exactly this order:

1. **Context** — why the equation is introduced at this point.
2. **Symbols** — every symbol: meaning, mathematical type, units, whether constant or variable.
3. **Assumptions** — every assumption, stated explicitly.
4. **Mathematical Background** — any mathematics not yet available is derived first.
5. **Step-by-Step Derivation** — every algebraic manipulation, without compression.
6. **Physical Interpretation** — what the equation means, and what each term does.
7. **Connection to the Next Equation** — how it leads into what follows.
8. **Consistency Checks** — limits, dimensions, numerical values.

Units

Both papers use the Gaussian (CGS) system, as was universal in magnetism in 1994. We use it throughout, without exception, so that every number in the papers can be reproduced directly:

Quantity	CGS unit	Meaning
length	cm	centimetre
mass	g	gram
time	s	second
force	dyn $= \mathrm{g\,cm\,s^{-2}}$	10^{-5} N
energy	erg $=$ dyn cm	10^{-7} J
dynamic viscosity	P (poise) $= \mathrm{g\,cm^{-1}s^{-1}}$	1 P $= 0.1$ Pa s; 1 cP $= 10^{-2}$ P
magnetic field H	Oe (oersted)	
magnetisation M, induction B	G (gauss)	1 G $\leftrightarrow$ 1 Oe in vacuum
magnetic moment m	$\mathrm{erg\,G^{-1}}$	
Boltzmann constant k	1.38×10^{-16} $\mathrm{erg\,K^{-1}}$	

A crucial convenience of Gaussian units: the energy of a magnetic moment m in a field H is $-mH$ with m in $\mathrm{erg\,G^{-1}}$ and H in Oe, so that the dimensionless ratio mH/kT contains no extra factors such as μ_0.

Typographic conventions

Vectors are bold upright-italic: $\boldsymbol{v}, \boldsymbol{H}, \boldsymbol{M}, \boldsymbol{\Omega}$. Their magnitudes are the same letter in italic: $v = |\boldsymbol{v}|$, $H = |\boldsymbol{H}|$. Unit vectors carry a hat, $\hat{\boldsymbol{h}} = \boldsymbol{H}/H$, except that we follow the papers in writing $\boldsymbol{h}$ (unbolded in the original) for the unit vector along the field; we will always write it $\hat{\boldsymbol{h}}$ to avoid ambiguity. Tensors are written in components, σ_{ij}. The symbol $\equiv$ means "is defined as".

Master Notation Table

Every symbol below is defined once, in the section indicated, and keeps that meaning in every volume of this book. Symbols not yet introduced in Volume I are listed here in advance so that the table can serve as a permanent reference.

Symbol	Meaning	Type	Units (CGS)
t	time	scalar, variable	s
$\boldsymbol{r}$, x_i	position vector / its components	vector	cm
$\boldsymbol{v}$	fluid velocity field	vector field	$\mathrm{cm\,s^{-1}}$
p	pressure	scalar field	$\mathrm{dyn\,cm^{-2}}$
ρ	mass density of the fluid	scalar	$\mathrm{g\,cm^{-3}}$
ρ_s	mass density of the solid particle material	scalar, constant	$\mathrm{g\,cm^{-3}}$
η	dynamic (shear) viscosity of the carrier liquid	scalar	P
$\nu = \eta/\rho$	kinematic viscosity	scalar	$\mathrm{cm^2\,s^{-1}}$
$\Delta\eta$	rotational (vortical) viscosity: the field-induced addition to η	scalar	P
$\boldsymbol{\Omega}$	local angular velocity of the fluid, $\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \boldsymbol{v}$	vector field	$\mathrm{s^{-1}}$
$\boldsymbol{\Theta}$, $\boldsymbol{\omega}_p$	macroscopic angular velocity of the particles ($\boldsymbol{\Theta}$ in SM, $\boldsymbol{\omega}_p$ in B)	vector field	$\mathrm{s^{-1}}$
d	particle diameter	scalar, constant	cm
$a = d/2$	particle radius	scalar, constant	cm
$V = \pi d^3/6$	volume of one particle	scalar, constant	$\mathrm{cm^3}$
n	number of particles per unit volume	scalar	$\mathrm{cm^{-3}}$
$\varphi = nV$	volume fraction of solid particles	scalar, dimensionless	—
I	total moment of inertia of the particles per unit volume	scalar	$\mathrm{g\,cm^{-1}}$
τ_s	inertial relaxation time of particle rotation	scalar	s
τ_B	Brownian rotational relaxation time	scalar	s
τ_h	hydrodynamic time scale l^2/ν	scalar	s
$\boldsymbol{H}$	magnetic field	vector field	Oe
H_0	amplitude of the alternating field	scalar, constant	Oe
ω	angular frequency of the field, $\omega = 2\pi f$	scalar, constant	$\mathrm{s^{-1}}$
f	frequency of the field	scalar	Hz
$\boldsymbol{M}$	magnetisation (magnetic moment per unit volume)	vector field	G
$\boldsymbol{M}_0$	instantaneous-equilibrium magnetisation	vector field	G

Symbol	Meaning	Type	Units (CGS)
M_s	spontaneous magnetisation of the particle material	scalar, constant	G
$m = M_s V$	magnetic moment of one particle	scalar, constant	$\mathrm{erg\,G^{-1}}$
$\boldsymbol{e}$	unit vector along a particle's magnetic moment	vector, variable	—
$\hat{\boldsymbol{h}}$	unit vector along $\boldsymbol{H}$	vector	—
$\xi = mH/kT$	Langevin parameter (field energy / thermal energy)	scalar, dimensionless	—
$\xi_0 = mH_0/kT$	Langevin parameter built on the field amplitude	scalar	—
ξ_e	effective-field Langevin parameter (Bacri et al.)	scalar	—
$L(\xi)$	Langevin function $\coth\xi - \xi^{-1}$	function	—
χ	initial (linear) magnetic susceptibility	scalar	—
kT	thermal energy	scalar	erg
T	absolute temperature	scalar	K
β	angle between $\boldsymbol{H}$ and $\boldsymbol{\Omega}$	scalar	rad
α	phase lag of $\boldsymbol{M}$ behind $\boldsymbol{H}$	scalar	rad
σ_{ij}	stress tensor	2nd-rank tensor	$\mathrm{dyn\,cm^{-2}}$
e_{ij}	rate-of-strain tensor	2nd-rank tensor	$\mathrm{s^{-1}}$
δ_{ij}	Kronecker delta	2nd-rank tensor	—
ε_{ijk}	Levi-Civita symbol	3rd-rank tensor	—
ζ_r	rotational friction coefficient of one particle	scalar	erg s
D_r	rotational diffusion coefficient	scalar	$\mathrm{s^{-1}}$
$W(\boldsymbol{e}, t)$	orientational distribution function	scalar field on the sphere	—
g, F, G, $\gamma(t)$	auxiliary functions of Secs. II–III of SM	functions	—

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Part I

Mathematical Foundations

Chapter 1

The Language of Vectors and Indices

1.1 Why we need a language before we need physics

Both papers are written in the language of three-dimensional vector calculus: their very first sentence contains $\boldsymbol{\Omega} = \nabla \times \boldsymbol{v}/2$, and their central object is a *cross product*, $\boldsymbol{M} \times \boldsymbol{H}$. None of this can be used until it has been built. This chapter builds it. Nothing in it is specific to ferrofluids; everything in it will be used constantly for the rest of the book.

1.2 Scalars, vectors and tensors

Definition 1.2.1 (Scalar)**.** A *scalar* is a quantity specified by a single number together with a unit, whose value does not change when the coordinate axes are rotated. Temperature T, pressure p and density ρ are scalars.

Definition 1.2.2 (Vector)**.** A *vector* is a quantity specified in three-dimensional space by three numbers (A_1, A_2, A_3) — its *components* along three mutually perpendicular axes — with the property that under a rotation of the axes the three numbers transform exactly as the components of a displacement do. Velocity $\boldsymbol{v}$, magnetic field $\boldsymbol{H}$ and angular velocity $\boldsymbol{\Omega}$ are vectors.

The transformation requirement is not decoration: it is what guarantees that an equation between vectors expresses a fact about nature and not about our choice of axes. Writing $\boldsymbol{A} = \boldsymbol{B}$ is the simultaneous statement of the three equations $A_1 = B_1$, $A_2 = B_2$, $A_3 = B_3$ in *every* coordinate system at once.

Definition 1.2.3 (Second-rank tensor)**.** A *second-rank tensor* is an object with two indices, T_{ij} $(i, j = 1, 2, 3)$, i.e. nine numbers, transforming under rotations as a product of two vector components does. The stress tensor σ_{ij} of Chapter 5 is the example we shall need.

Physically, a second-rank tensor is what one needs whenever a vector cause produces a vector effect that is *not* parallel to it. A force per unit area acting on a surface depends on the orientation of that surface: two directions are involved, hence two indices.

1.3 Index notation and the summation convention

Writing $\boldsymbol{A} \cdot \boldsymbol{B} = A_1B_1 + A_2B_2 + A_3B_3$ becomes intolerable in long calculations. We therefore adopt *index notation*: a vector $\boldsymbol{A}$ is represented by its typical component A_i, where the free index i silently runs over $1, 2, 3$.

Definition 1.3.1 (Einstein summation convention)**.** Whenever an index appears *exactly twice* in a single product, a sum over $1, 2, 3$ on that index is understood, and the summation sign is omitted:

$$A_i B_i \equiv \sum_{i=1}^{3} A_i B_i = A_1 B_1 + A_2 B_2 + A_3 B_3. \tag{1.3.1}$$

A twice-repeated index is called a *dummy* (or bound) index; a once-appearing index is a *free* index.

Two rules follow immediately and will be used without further comment:

1. A dummy index may be renamed at will: $A_i B_i = A_k B_k$, because both are the same sum of three terms.
2. Every term of a valid equation must carry the *same* free indices. An equation $A_i = B_{ij} C_j$ is meaningful (free index i on both sides, j summed); an equation $A_i = B_j$ is not.

Thus the scalar product is

$$\boldsymbol{A} \cdot \boldsymbol{B} = A_i B_i, \tag{1.3.2}$$

a scalar (no free index), and the squared length of a vector is $A^2 = A_i A_i$.

1.4 The Kronecker delta

Definition 1.4.1 (Kronecker delta)**.**

$$\delta_{ij} \equiv \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \tag{1.4.1}$$

Written as a matrix, δ_{ij} is the 3×3 identity. Its single most used property is the *substitution property*: for any indexed quantity A_j,

$$\delta_{ij} A_j = \delta_{i1} A_1 + \delta_{i2} A_2 + \delta_{i3} A_3 = A_i, \tag{1.4.2}$$

because exactly one of the three deltas equals 1 (the one with $j = i$) and the other two vanish. In words: *contracting an index with a Kronecker delta renames that index.* Two further consequences, obtained the same way:

$$\delta_{ii} = \delta_{11} + \delta_{22} + \delta_{33} = 3, \tag{1.4.3}$$

$$\delta_{ij} \delta_{jk} = \delta_{ik}. \tag{1.4.4}$$

The first is the trace of the identity in three dimensions; the second follows by applying (1.4.2) with $A_j \to \delta_{jk}$.

1.5 The Levi-Civita symbol and the cross product

Definition 1.5.1 (Levi-Civita symbol)**.**

$$\varepsilon_{ijk} \equiv \begin{cases} +1, & (ijk) \text{ an even (cyclic) permutation of } (123) : \ 123,\ 231,\ 312, \\ -1, & (ijk) \text{ an odd permutation of } (123) : \ 132,\ 321,\ 213, \\ 0, & \text{any two indices equal.} \end{cases} \tag{1.5.1}$$

Its defining property is *total antisymmetry*: interchanging any two indices changes the sign,

$$\varepsilon_{ijk} = -\varepsilon_{jik} = -\varepsilon_{ikj} = -\varepsilon_{kji}, \qquad \varepsilon_{ijk} = \varepsilon_{jki} = \varepsilon_{kij}. \tag{1.5.2}$$

The second line says a *cyclic* shift (two interchanges) leaves the sign unchanged. The vanishing of ε_{ijk} when two indices coincide is forced by antisymmetry: if $\varepsilon_{iik} = -\varepsilon_{iik}$, then $\varepsilon_{iik} = 0$.

Definition 1.5.2 (Cross product)**.**

$$(\boldsymbol{A} \times \boldsymbol{B})_i \equiv \varepsilon_{ijk} A_j B_k. \tag{1.5.3}$$

Let us check that this reproduces the familiar formula. Take $i = 1$. The sum over j, k contains only terms with $\varepsilon_{1jk} \neq 0$, i.e. $(j, k) = (2, 3)$ giving $\varepsilon_{123} = +1$ and $(j, k) = (3, 2)$ giving $\varepsilon_{132} = -1$:

$$(\boldsymbol{A} \times \boldsymbol{B})_1 = \varepsilon_{123} A_2 B_3 + \varepsilon_{132} A_3 B_2 = A_2 B_3 - A_3 B_2, \tag{1.5.4}$$

which is indeed the first component of the elementary cross product. The other two components follow by cyclic permutation, $1 \to 2 \to 3 \to 1$.

Two properties follow at once from antisymmetry (1.5.2):

$$(\boldsymbol{A} \times \boldsymbol{B})_i = \varepsilon_{ijk} A_j B_k = -\varepsilon_{ikj} A_j B_k = -\varepsilon_{ijk} A_k B_j = -(\boldsymbol{B} \times \boldsymbol{A})_i, \tag{1.5.5}$$

$$\boldsymbol{A} \times \boldsymbol{A} = \boldsymbol{0}, \tag{1.5.6}$$

where in (1.5.5) we first used the antisymmetry of ε in its last two indices, then *renamed the dummy indices* $j \leftrightarrow k$ (legal, because both are summed).

Also, contracting a cross product with one of its own factors gives zero:

$$\boldsymbol{A} \cdot (\boldsymbol{A} \times \boldsymbol{B}) = A_i \varepsilon_{ijk} A_j B_k = 0, \tag{1.5.7}$$

because ε_{ijk} is antisymmetric in $i \leftrightarrow j$ while $A_i A_j$ is symmetric, and the contraction of a symmetric with an antisymmetric pair of indices always vanishes. (Proof of that statement: if $S_{ij} = S_{ji}$ and $A_{ij} = -A_{ji}$, then $S_{ij} A_{ij} = S_{ji} A_{ij} = -S_{ji} A_{ji} = -S_{ij} A_{ij}$ after renaming, so $S_{ij} A_{ij} = 0$.) Equation (1.5.7) is the statement that $\boldsymbol{A} \times \boldsymbol{B}$ is perpendicular to $\boldsymbol{A}$, and by the same argument to $\boldsymbol{B}$: the cross product points along the normal to the plane of the two vectors. This is why torques and angular velocities — quantities that characterise *rotation in a plane* — are represented by vectors along the rotation axis.

1.6 The ε–δ identity

The single most useful algebraic identity in this book is

$$\boxed{\varepsilon_{ijk} \varepsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}} \tag{1.6.1}$$

(the repeated index k is summed).

Proof. Both sides are numbers determined by the four free indices i, j, l, m, so we compare them case by case.

Case 1: $i = j$. The left side contains $\varepsilon_{ijk} = \varepsilon_{iik} = 0$, so it vanishes. The right side becomes $\delta_{il}\delta_{im} - \delta_{im}\delta_{il} = 0$. Both sides vanish. The same argument applies if $l = m$.

Case 2: $i \neq j$ and $l \neq m$. For fixed $i \neq j$, the sum over k has exactly one nonzero term, namely the k that differs from both i and j; similarly ε_{klm} is nonzero only for the k differing from both l and m. Hence the left side is nonzero only if $\{i, j\} = \{l, m\}$ as sets.

Case 2a: $i = l$, $j = m$ (with $i \neq j$). Then the left side is $\varepsilon_{ijk}\varepsilon_{kij} = \varepsilon_{ijk}\varepsilon_{ijk}$ (cyclic shift), which is $(\pm 1)^2 = 1$ for the single surviving k. The right side is $\delta_{ii}^{(\text{no sum})}\delta_{jj}^{(\text{no sum})} - \delta_{ij}\delta_{ji} = 1 \cdot 1 - 0 = 1$. Equal.

Case 2b: $i = m$, $j = l$ (with $i \neq j$). The left side is $\varepsilon_{ijk}\varepsilon_{kji} = -\varepsilon_{ijk}\varepsilon_{kij} = -1$ by Case 2a and one interchange. The right side is $\delta_{ij}\delta_{ji} - \delta_{ii}\delta_{jj} = 0 - 1 = -1$. Equal.

Case 2c: $\{i, j\} \neq \{l, m\}$. The left side vanishes, as argued. On the right, $\delta_{il}\delta_{jm}$ and $\delta_{im}\delta_{jl}$ both vanish, since each requires $\{i, j\} = \{l, m\}$. Equal.

All cases agree, so (1.6.1) holds identically. ■

1.7 Triple products

Result 1.7.1 (Vector triple product)**.**

$$\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \boldsymbol{B}\,(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}\,(\boldsymbol{A} \cdot \boldsymbol{B}). \tag{1.7.1}$$

Derivation. Write the ith component using (1.5.3) twice. The inner product $(\boldsymbol{B} \times \boldsymbol{C})_k = \varepsilon_{klm}B_lC_m$, hence

$$\begin{aligned}
[\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C})]_i &= \varepsilon_{ijk}A_j(\boldsymbol{B} \times \boldsymbol{C})_k && && (1.7.2)\\
&= \varepsilon_{ijk}A_j\,\varepsilon_{klm}B_lC_m && && (1.7.3)\\
&= \varepsilon_{ijk}\varepsilon_{klm}\,A_jB_lC_m && && (1.7.4)\\
&= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})A_jB_lC_m && \text{by (1.6.1)} && (1.7.5)\\
&= \delta_{il}\delta_{jm}A_jB_lC_m - \delta_{im}\delta_{jl}A_jB_lC_m && && (1.7.6)\\
&= A_mB_iC_m - A_lB_lC_i && \text{by the substitution property (1.4.2)} && (1.7.7)\\
&= B_i(A_mC_m) - C_i(A_lB_l) && && (1.7.8)\\
&= B_i(\boldsymbol{A} \cdot \boldsymbol{C}) - C_i(\boldsymbol{A} \cdot \boldsymbol{B}). && && (1.7.9)
\end{aligned}$$

Since this holds for every i, (1.7.1) follows. ■

Equation (1.7.1) is used in this book more than any other identity; in particular the combination $(\boldsymbol{e} \times \boldsymbol{H}) \times \boldsymbol{e}$, which converts a magnetic torque into a magnetic "force" on the orientation of a particle, is evaluated with it in Volume II.

Result 1.7.2 (Scalar triple product)**.**

$$\boldsymbol{A} \cdot (\boldsymbol{B} \times \boldsymbol{C}) = \varepsilon_{ijk}A_iB_jC_k = \boldsymbol{B} \cdot (\boldsymbol{C} \times \boldsymbol{A}) = \boldsymbol{C} \cdot (\boldsymbol{A} \times \boldsymbol{B}). \tag{1.7.10}$$

Derivation. $\boldsymbol{A} \cdot (\boldsymbol{B} \times \boldsymbol{C}) = A_i\varepsilon_{ijk}B_jC_k = \varepsilon_{ijk}A_iB_jC_k$. Now use the cyclic invariance $\varepsilon_{ijk} = \varepsilon_{jki}$ from (1.5.2): $\varepsilon_{ijk}A_iB_jC_k = \varepsilon_{jki}A_iB_jC_k = B_j\varepsilon_{jki}C_kA_i = \boldsymbol{B} \cdot (\boldsymbol{C} \times \boldsymbol{A})$, and once more for the third form. ■

This will be needed when a torque is projected onto the vorticity axis: the combination $\boldsymbol{\Omega} \cdot (\boldsymbol{e} \times \hat{\boldsymbol{h}})$ appears in the rotational diffusion problem of Volume II, and (1.7.10) lets us move the factors around freely.

Chapter 2

Vector Calculus

2.1 Fields

A *field* is a quantity defined at every point of space and at every instant. A *scalar field* $p(\boldsymbol{r}, t)$ assigns a number to each point (e.g. pressure); a *vector field* $\boldsymbol{v}(\boldsymbol{r}, t)$ assigns a vector (e.g. velocity). Continuum physics is entirely a theory of fields: instead of tracking 10^{20} molecules we track a handful of smooth functions.

Partial derivatives are abbreviated

$$\partial_i \equiv \frac{\partial}{\partial x_i}, \tag{2.1.1}$$

so that $\partial_i x_j = \delta_{ij}$: the coordinates are independent variables, and $\partial x_1/\partial x_2 = 0$ while $\partial x_1/\partial x_1 = 1$. This tiny identity is used repeatedly in Chapter 6.

2.2 Gradient

Definition 2.2.1 (Gradient)**.** For a scalar field f, $(\nabla f)_i \equiv \partial_i f$.

Why this is the right object. Consider moving by a small displacement $\mathrm{d}\boldsymbol{r}$ with components $\mathrm{d}x_i$. The change in f is, by the chain rule for functions of several variables,

$$\mathrm{d}f = \frac{\partial f}{\partial x_1}\mathrm{d}x_1 + \frac{\partial f}{\partial x_2}\mathrm{d}x_2 + \frac{\partial f}{\partial x_3}\mathrm{d}x_3 = \partial_i f \,\mathrm{d}x_i = \nabla f \cdot \mathrm{d}\boldsymbol{r}. \tag{2.2.1}$$

Thus ∇f is the unique vector whose scalar product with any displacement gives the resulting change of f. Writing $\nabla f \cdot \mathrm{d}\boldsymbol{r} = |\nabla f|\,|\mathrm{d}\boldsymbol{r}|\cos\theta$, the change is largest when $\theta = 0$: *the gradient points in the direction of steepest increase, and its magnitude is the rate of increase in that direction.* Its units are those of f divided by cm.

2.3 Divergence and the divergence theorem

Definition 2.3.1 (Divergence)**.** For a vector field $\boldsymbol{A}$, $\nabla \cdot \boldsymbol{A} \equiv \partial_i A_i$, a scalar field.

Why this is the right object. Take a small rectangular box with one corner at $\boldsymbol{r}$ and edges $\delta x_1, \delta x_2, \delta x_3$ parallel to the axes. Compute the net outward flux of $\boldsymbol{A}$ through its surface — "how much of $\boldsymbol{A}$ leaves the box per unit time". Through the face at $x_1 + \delta x_1$ (outward normal

$+\hat{\boldsymbol{x}}_1$, area $\delta x_2 \delta x_3$) the flux is $A_1(x_1+\delta x_1, x_2, x_3)\,\delta x_2\delta x_3$; through the opposite face (outward normal $-\hat{\boldsymbol{x}}_1$) it is $-A_1(x_1,x_2,x_3)\,\delta x_2\delta x_3$. The sum is

$$\big[A_1(x_1+\delta x_1)-A_1(x_1)\big]\delta x_2\delta x_3 = \left[\frac{\partial A_1}{\partial x_1}\delta x_1 + O(\delta x_1^2)\right]\delta x_2\delta x_3 \tag{2.3.1}$$

$$\approx \frac{\partial A_1}{\partial x_1}\,\delta x_1\delta x_2\delta x_3. \tag{2.3.2}$$

The Taylor expansion in (2.3.1) is legitimate because A_1 is assumed differentiable and δx_1 is arbitrarily small; the neglected term is smaller than the retained one by a factor δx_1, which we let tend to zero. Adding the two remaining pairs of faces and dividing by the volume $\delta V = \delta x_1\delta x_2\delta x_3$,

$$\frac{\text{net outward flux}}{\delta V} = \partial_1 A_1 + \partial_2 A_2 + \partial_3 A_3 = \nabla\cdot\boldsymbol{A}. \tag{2.3.3}$$

So the divergence is the *outflow per unit volume*: a source (>0), a sink (<0), or neither ($=0$).

Result 2.3.1 (Divergence (Gauss) theorem)**.** *For a volume $\mathcal{V}$ bounded by a closed surface $\mathcal{S}$ with outward unit normal $\boldsymbol{n}$,*

$$\int_{\mathcal{V}} (\nabla\cdot\boldsymbol{A})\,\mathrm{d}V = \oint_{\mathcal{S}} \boldsymbol{A}\cdot\boldsymbol{n}\,\mathrm{d}S. \tag{2.3.4}$$

Derivation. Chop $\mathcal{V}$ into many tiny boxes. By (2.3.3), for each box the outward flux equals $(\nabla\cdot\boldsymbol{A})\delta V$. Sum over all boxes. Every internal face is shared by two neighbouring boxes and carries opposite outward normals for them, so the two contributions cancel exactly. Only the faces lying on the outer boundary $\mathcal{S}$ survive. Hence the sum of $(\nabla\cdot\boldsymbol{A})\delta V$ over the interior equals the flux through $\mathcal{S}$; in the limit of infinitely fine subdivision the sums become the integrals in (2.3.4). ■

This theorem is what converts a statement about a *surface* (fluxes, forces on a boundary) into a statement about a *volume* (local differential equations). It is the engine of Chapter 5.

2.4 Curl

Definition 2.4.1 (Curl)**.** $(\nabla\times\boldsymbol{A})_i \equiv \varepsilon_{ijk}\partial_j A_k$.

Why this is the right object. Consider a small rectangle in the x_1x_2 plane with corners (x_1,x_2), $(x_1+\delta x_1, x_2)$, $(x_1+\delta x_1, x_2+\delta x_2)$, $(x_1, x_2+\delta x_2)$, and compute the *circulation* $\oint \boldsymbol{A}\cdot\mathrm{d}\boldsymbol{l}$ counterclockwise:

$$\oint \boldsymbol{A}\cdot\mathrm{d}\boldsymbol{l} = \underbrace{A_1(x_1,x_2)\,\delta x_1}_{\text{bottom}} + \underbrace{A_2(x_1+\delta x_1,x_2)\,\delta x_2}_{\text{right}} - \underbrace{A_1(x_1,x_2+\delta x_2)\,\delta x_1}_{\text{top}} - \underbrace{A_2(x_1,x_2)\,\delta x_2}_{\text{left}} \tag{2.4.1}$$

$$= \big[A_2(x_1+\delta x_1,x_2)-A_2(x_1,x_2)\big]\delta x_2 - \big[A_1(x_1,x_2+\delta x_2)-A_1(x_1,x_2)\big]\delta x_1 \tag{2.4.2}$$

$$= \left(\frac{\partial A_2}{\partial x_1}\delta x_1\right)\delta x_2 - \left(\frac{\partial A_1}{\partial x_2}\delta x_2\right)\delta x_1 + O(\delta^3) \tag{2.4.3}$$

$$= (\partial_1 A_2 - \partial_2 A_1)\,\delta x_1\delta x_2. \tag{2.4.4}$$

The signs of the top and left segments are negative because the path traverses them in the direction opposite to the coordinate axis. Dividing by the enclosed area $\delta x_1\delta x_2$,

$$\frac{\text{circulation}}{\text{area}} = \partial_1 A_2 - \partial_2 A_1 = (\nabla\times\boldsymbol{A})_3. \tag{2.4.5}$$

Thus the component of the curl along a given axis is the circulation per unit area in the plane perpendicular to that axis: *the curl measures local rotation.* That is precisely why $\nabla \times \boldsymbol{v}$ will turn out to be twice the local angular velocity of the fluid (Chapter 3), which is the content of the very first formula of the Shliomis–Morozov paper.

The same box-and-cancellation argument as before converts (2.4.5) into *Stokes' theorem*,

$$\int_{\mathcal{S}} (\nabla \times \boldsymbol{A}) \cdot \boldsymbol{n} \, \mathrm{d}S = \oint_{\partial \mathcal{S}} \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{l}, \tag{2.4.6}$$

where now the interior edges shared by adjacent small loops are traversed twice in opposite directions and cancel.

2.5 Laplacian

Definition 2.5.1 (Laplacian)**.** $\nabla^2 f \equiv \nabla \cdot (\nabla f) = \partial_i \partial_i f$, and for a vector field, componentwise, $(\nabla^2 \boldsymbol{A})_i = \partial_j \partial_j A_i$.

Its meaning: $\nabla^2 f$ at a point is proportional to the amount by which the average of f over a small sphere around the point exceeds f at the point itself. To see this in one line for one dimension, expand $f(x \pm \delta) = f \pm f'\delta + \tfrac{1}{2} f''\delta^2 + \dots$ and add:

$$\tfrac{1}{2}\big[f(x+\delta) + f(x-\delta)\big] - f(x) = \tfrac{1}{2} f''(x)\delta^2. \tag{2.5.1}$$

A positive Laplacian means the neighbourhood is "higher" than the point, so diffusion-like processes ($\partial_t f = D\nabla^2 f$) push f up there. In the Navier–Stokes equation the term $\eta \nabla^2 \boldsymbol{v}$ plays exactly this role for momentum: viscosity diffuses momentum from fast to slow layers.

2.6 Two identities we shall need

Result 2.6.1.

$$\nabla \times (\nabla \times \boldsymbol{A}) = \nabla(\nabla \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A}. \tag{2.6.1}$$

Derivation.

$$\begin{aligned}
[\nabla \times (\nabla \times \boldsymbol{A})]_i &= \varepsilon_{ijk} \partial_j (\nabla \times \boldsymbol{A})_k = \varepsilon_{ijk} \partial_j \big(\varepsilon_{klm} \partial_l A_m\big) && (2.6.2)\\
&= \varepsilon_{ijk}\varepsilon_{klm} \, \partial_j \partial_l A_m && (2.6.3)\\
&= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) \partial_j \partial_l A_m && (2.6.4)\\
&= \partial_j \partial_i A_j - \partial_j \partial_j A_i && (2.6.5)\\
&= \partial_i (\partial_j A_j) - \partial_j \partial_j A_i && (2.6.6)\\
&= [\nabla(\nabla \cdot \boldsymbol{A})]_i - (\nabla^2 \boldsymbol{A})_i. && (2.6.7)
\end{aligned}$$

In the fifth line the order of the partial derivatives was exchanged, $\partial_j \partial_i = \partial_i \partial_j$, which is permitted for fields with continuous second derivatives (Clairaut's theorem). ∎

Result 2.6.2.

$$\nabla \cdot (\nabla \times \boldsymbol{A}) = 0. \tag{2.6.8}$$

Derivation. $\partial_i(\varepsilon_{ijk}\partial_j A_k) = \varepsilon_{ijk}\partial_i\partial_j A_k$. The factor $\partial_i\partial_j$ is symmetric under $i \leftrightarrow j$ while ε_{ijk} is antisymmetric; by the symmetric–antisymmetric lemma proved after (1.5.7), the contraction vanishes. ■

Identity (2.6.8) guarantees that any velocity field written as a curl is automatically incompressible — a fact we exploit in Chapter 6 to write down the flow around a rotating sphere without solving a differential equation.

Chapter 3

Kinematics: How a Fluid Element Moves, Deforms and Spins

3.1 Why this chapter exists

The opening sentence of Shliomis & Morozov reads: "Moving with a velocity $\boldsymbol{v}$, each element of the liquid volume rotates with a local angular velocity $\boldsymbol{\Omega} = \nabla \times \boldsymbol{v}/2$." That statement is not a definition plucked from the air; it is a theorem about the local structure of an arbitrary velocity field. This chapter proves it. Without it, nothing in either paper — which is *entirely* about the difference between the spin of the particles and the spin of the fluid — can even be formulated.

3.2 The material derivative

Let $f(\boldsymbol{r}, t)$ be any property carried by the fluid. A fluid element that is at $\boldsymbol{r}$ at time t is at $\boldsymbol{r} + \boldsymbol{v}\,\mathrm{d}t$ at time $t + \mathrm{d}t$. The change of f *following the element* is therefore

$$\mathrm{d}f = f(\boldsymbol{r} + \boldsymbol{v}\,\mathrm{d}t,\ t + \mathrm{d}t) - f(\boldsymbol{r}, t) \tag{3.2.1}$$

$$= \frac{\partial f}{\partial t}\mathrm{d}t + \frac{\partial f}{\partial x_i}\, v_i\,\mathrm{d}t + O(\mathrm{d}t^2), \tag{3.2.2}$$

using the multivariable Taylor expansion (legitimate for differentiable f and infinitesimal $\mathrm{d}t$). Dividing by $\mathrm{d}t$ and letting $\mathrm{d}t \to 0$,

$$\boxed{\frac{\mathrm{D}f}{\mathrm{D}t} \equiv \frac{\partial f}{\partial t} + (\boldsymbol{v}\cdot\nabla)f} \tag{3.2.3}$$

This is the *material* (substantial, convective) derivative. The first term is the change at a fixed point in space; the second is the change caused by the element being swept into a region where f has a different value. In both papers the symbol $\mathrm{d}/\mathrm{d}t$ in the equations of motion always means (3.2.3); Shliomis & Morozov state this explicitly below their Eq. (SM-26), writing $\mathrm{d}/\mathrm{d}t = \partial/\partial t + (\boldsymbol{v}_0\cdot\nabla)$.

3.3 Local decomposition of a velocity field

Fix a point $\boldsymbol{r}$ and look at the velocity a small distance $\boldsymbol{\delta}$ away. Taylor expansion (each component separately) gives

$$v_i(\boldsymbol{r}+\boldsymbol{\delta}) = v_i(\boldsymbol{r}) + \big(\partial_j v_i\big)\delta_j + O(\delta^2). \tag{3.3.1}$$

The three terms will turn out to describe, in order: translation, deformation, and rotation. To separate the last two, split the velocity-gradient tensor $\partial_j v_i$ into symmetric and antisymmetric parts. This is always possible and always unique:

$$\partial_j v_i = \underbrace{\tfrac{1}{2}\big(\partial_j v_i + \partial_i v_j\big)}_{e_{ij}} + \underbrace{\tfrac{1}{2}\big(\partial_j v_i - \partial_i v_j\big)}_{a_{ij}}. \tag{3.3.2}$$

That the identity holds is checked by adding the two brackets: the terms $\frac{1}{2}\partial_i v_j$ cancel and $2\times\frac{1}{2}\partial_j v_i$ remains. Uniqueness follows because $e_{ij}=e_{ji}$ and $a_{ij}=-a_{ji}$, and a tensor that is both symmetric and antisymmetric is zero.

Definition 3.3.1 (Rate-of-strain tensor)**.** $e_{ij}\equiv\frac{1}{2}(\partial_j v_i+\partial_i v_j)$, units s^{-1}. It measures the rate at which the element is stretched and sheared.

3.4 The antisymmetric part is a rigid rotation

Result 3.4.1. *The antisymmetric part of the velocity gradient acts on $\boldsymbol{\delta}$ exactly as a rigid-body rotation with angular velocity*

$$\boxed{\boldsymbol{\Omega} = \tfrac{1}{2}\nabla\times\boldsymbol{v}} \tag{3.4.1}$$

that is, $a_{ij}\delta_j = (\boldsymbol{\Omega}\times\boldsymbol{\delta})_i$.

Step 1: what rigid rotation looks like. A rigid body rotating with angular velocity $\boldsymbol{\Omega}$ about a point has, at displacement $\boldsymbol{\delta}$ from that point, the velocity $\boldsymbol{\Omega}\times\boldsymbol{\delta}$. (Reason: the point moves on a circle of radius $\delta\sin\theta$ about the axis, where θ is the angle between $\boldsymbol{\Omega}$ and $\boldsymbol{\delta}$; its speed is $\Omega\,\delta\sin\theta = |\boldsymbol{\Omega}\times\boldsymbol{\delta}|$, and its direction is perpendicular to both the axis and $\boldsymbol{\delta}$, which is the direction of $\boldsymbol{\Omega}\times\boldsymbol{\delta}$ by (1.5.7).)

Step 2: evaluate $\boldsymbol{\Omega}\times\boldsymbol{\delta}$ with $\boldsymbol{\Omega}$ given by (3.4.1)**.** In components, using (1.5.3) and the definition of the curl,

$$\begin{aligned}(\boldsymbol{\Omega}\times\boldsymbol{\delta})_i &= \varepsilon_{ijk}\Omega_j\delta_k && (3.4.2)\\ &= \varepsilon_{ijk}\left(\tfrac{1}{2}\varepsilon_{jpq}\partial_p v_q\right)\delta_k && (3.4.3)\\ &= \tfrac{1}{2}\,\varepsilon_{ijk}\varepsilon_{jpq}\,\partial_p v_q\,\delta_k. && (3.4.4)\end{aligned}$$

To apply the ε–δ identity (1.6.1) we need the summed index (j) to sit in the *last* slot of the first symbol and the *first* slot of the second. Use cyclic invariance (1.5.2) on the first factor: $\varepsilon_{ijk}=\varepsilon_{jki}$. Then

$$\begin{aligned}\tfrac{1}{2}\,\varepsilon_{jki}\varepsilon_{jpq}\,\partial_p v_q\,\delta_k &= \tfrac{1}{2}\,\varepsilon_{kij}\varepsilon_{jpq}\,\partial_p v_q\,\delta_k && \text{(cyclic again: } \varepsilon_{jki}=\varepsilon_{kij}) && (3.4.5)\\ &= \tfrac{1}{2}\big(\delta_{kp}\delta_{iq}-\delta_{kq}\delta_{ip}\big)\partial_p v_q\,\delta_k && \text{by (1.6.1) with } (i,j,k,l,m)\to(k,i,j,p,q) && (3.4.6)\end{aligned}$$

$$= \tfrac{1}{2}\Big(\delta_{kp}\delta_{iq}\partial_p v_q \delta_k - \delta_{kq}\delta_{ip}\partial_p v_q \delta_k\Big) \tag{3.4.7}$$

$$= \tfrac{1}{2}\Big(\partial_k v_i\, \delta_k - \partial_i v_k\, \delta_k\Big) \qquad \text{substitution property (1.4.2)} \tag{3.4.8}$$

$$= \tfrac{1}{2}\big(\partial_k v_i - \partial_i v_k\big)\delta_k \tag{3.4.9}$$

$$= a_{ik}\delta_k, \tag{3.4.10}$$

which is exactly the antisymmetric part of (3.3.2) contracted with $\boldsymbol{\delta}$ (the dummy index is called k instead of j, which is allowed). ■

Step 3: the complete local picture. Substituting (3.3.2) into (3.3.1),

$$v_i(\boldsymbol{r}+\boldsymbol{\delta}) = \underbrace{v_i(\boldsymbol{r})}_{\text{translation}} + \underbrace{e_{ij}\delta_j}_{\text{deformation}} + \underbrace{(\boldsymbol{\Omega}\times\boldsymbol{\delta})_i}_{\text{rigid rotation}} . \tag{3.4.11}$$

Every flow, however complicated, looks locally like a translation plus a stretching plus a rigid spin. A small solid sphere suspended in the fluid cannot be stretched (it is rigid), and it is carried along by the translation; what it *can* do is spin. In the absence of any other torque it simply spins with the fluid, at the rate $\boldsymbol{\Omega}$. This is the physical content of the statement $\boldsymbol{\Theta} = \boldsymbol{\Omega}$ that opens the Shliomis–Morozov paper, and the whole of ferrofluid magnetoviscosity consists of making $\boldsymbol{\Theta} \neq \boldsymbol{\Omega}$.

Remark 3.4.1. The factor $\frac{1}{2}$ in (3.4.1) is not a convention: it is forced. The curl measures circulation per unit area, and for a rigid rotation $\boldsymbol{v} = \boldsymbol{\Omega}\times\boldsymbol{r}$ one finds (Exercise, done here for completeness)

$$(\nabla\times\boldsymbol{v})_i = \varepsilon_{ijk}\partial_j\big(\varepsilon_{kpq}\Omega_p x_q\big) = \varepsilon_{ijk}\varepsilon_{kpq}\Omega_p\,\partial_j x_q \tag{3.4.12}$$

$$= \varepsilon_{ijk}\varepsilon_{kpq}\Omega_p\delta_{jq} = (\delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp})\Omega_p\delta_{jq} \tag{3.4.13}$$

$$= \delta_{ip}\Omega_p\delta_{jj} - \delta_{jp}\Omega_p\delta_{ij} = 3\Omega_i - \Omega_i = 2\Omega_i. \tag{3.4.14}$$

So the curl of a rigid rotation is *twice* the angular velocity, and (3.4.1) inverts this.

3.5 Worked example: vorticity in the Poiseuille flow of the experiment

Bacri *et al.* measure viscosity in a horizontal capillary of diameter $d_{\mathrm{tube}} = 1$ mm through which the ferrofluid is driven by a pressure difference. The flow is a cylindrical Poiseuille flow: with z along the tube axis and r the distance from it,

$$v_z(r) = \frac{\Delta p}{4\eta\ell}\big(R^2 - r^2\big), \qquad R = d_{\mathrm{tube}}/2, \tag{3.5.1}$$

ℓ being the tube length and Δp the pressure drop. (This profile is *derived* in Volume II once the Navier–Stokes equation is available; here we only need its shape.) The velocity has a single component depending on a single coordinate, so the curl has one surviving component, along the azimuthal direction $\hat{\boldsymbol{\phi}}$:

$$\nabla\times\boldsymbol{v} = -\frac{\mathrm{d}v_z}{\mathrm{d}r}\,\hat{\boldsymbol{\phi}}, \qquad \boldsymbol{\Omega} = -\frac{1}{2}\frac{\mathrm{d}v_z}{\mathrm{d}r}\,\hat{\boldsymbol{\phi}} = \frac{\Delta p\, r}{4\eta\ell}\,\hat{\boldsymbol{\phi}}. \tag{3.5.2}$$

Three facts, all of which matter later:

1. $\boldsymbol{\Omega}$ is *azimuthal*: it is perpendicular to the tube axis. The solenoid in the experiment produces $\boldsymbol{H}$ *along* the tube axis. Hence $\boldsymbol{H} \perp \boldsymbol{\Omega}$, i.e. the angle between them is $\beta = \pi/2$ — exactly the case assumed in Eq. (SM-2) and the reason Shliomis & Morozov write "or in a cylindrical Poiseuille flow when the capillary tube is aligned along the axis of the magnetic coil".

2. $\boldsymbol{\Omega}$ vanishes on the axis and is maximal at the wall. The measured viscosity is therefore an average over a distribution of local vorticities; this is why a *Newtonian* response (flow rate $\propto$ pressure drop, their Fig. 1) is a nontrivial experimental check.

3. Numerically, Bacri *et al.* state that $\Omega < 60\ \mathrm{s}^{-1}$ throughout their experiments. With $\tau_B \simeq 1.6$ ms this gives $\Omega\tau_B < 60 \times 1.6 \times 10^{-3} \approx 0.1 \ll 1$, which is the small parameter that makes the entire perturbation theory of both papers legitimate.

Part II

Physical Foundations

Chapter 4

Mechanics of Rotating Bodies

4.1 Newton's laws, stated and used

Assumption 4.1.1 (Newton's laws)**.** (i) In an inertial frame a body with no net force on it moves with constant velocity. (ii) $\boldsymbol{F} = \mathrm{d}\boldsymbol{P}/\mathrm{d}t$ with $\boldsymbol{P} = m_p\boldsymbol{u}$ the linear momentum. (iii) Forces between two bodies are equal and opposite.

These are empirical postulates; they cannot be derived from anything more primitive, and we take them as the starting point of all mechanics in this book. Everything else in this chapter *is* derived.

4.2 Angular momentum and torque

Definition 4.2.1 (Angular momentum and torque)**.** For a point mass m_p at position $\boldsymbol{r}$ (measured from a chosen origin) moving with velocity $\boldsymbol{u}$,

$$\boldsymbol{L} \equiv \boldsymbol{r} \times m_p\boldsymbol{u} = \boldsymbol{r} \times \boldsymbol{P}, \qquad \boldsymbol{T} \equiv \boldsymbol{r} \times \boldsymbol{F}. \tag{4.2.1}$$

$\boldsymbol{L}$ has units $\mathrm{g\,cm^2 s^{-1}} = \mathrm{erg\,s}$; $\boldsymbol{T}$ has units $\mathrm{dyn\,cm} = \mathrm{erg}$.

Result 4.2.1.

$$\frac{\mathrm{d}\boldsymbol{L}}{\mathrm{d}t} = \boldsymbol{T}. \tag{4.2.2}$$

Derivation. Differentiate the definition with the product rule (valid componentwise, since $(\boldsymbol{A}\times\boldsymbol{B})_i = \varepsilon_{ijk}A_jB_k$ is bilinear):

$$\frac{\mathrm{d}\boldsymbol{L}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\big(\boldsymbol{r}\times\boldsymbol{P}\big) = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t}\times\boldsymbol{P} + \boldsymbol{r}\times\frac{\mathrm{d}\boldsymbol{P}}{\mathrm{d}t} \tag{4.2.3}$$

$$= \boldsymbol{u}\times m_p\boldsymbol{u} + \boldsymbol{r}\times\boldsymbol{F} \qquad \text{(Newton II for the second term)} \tag{4.2.4}$$

$$= \boldsymbol{0} + \boldsymbol{T}, \tag{4.2.5}$$

because $\boldsymbol{u}\times\boldsymbol{u} = \boldsymbol{0}$. ■

Equation (4.2.2) is the rotational analogue of $\boldsymbol{F} = \mathrm{d}\boldsymbol{P}/\mathrm{d}t$ and is the equation obeyed by every particle in this book. For a *rigid body* rotating with angular velocity $\boldsymbol{\theta}$ about a symmetry axis it becomes $I\,\mathrm{d}\boldsymbol{\theta}/\mathrm{d}t = \boldsymbol{T}$, where I is the moment of inertia, defined next.

4.3 Moment of inertia, and its value for a sphere

For a rigid body rotating with angular velocity $\boldsymbol{\theta}$, each mass element $\mathrm{d}m$ at position $\boldsymbol{r}$ from the rotation axis moves with $\boldsymbol{u} = \boldsymbol{\theta} \times \boldsymbol{r}$. Its angular momentum is $\mathrm{d}\boldsymbol{L} = \boldsymbol{r} \times (\boldsymbol{\theta} \times \boldsymbol{r})\mathrm{d}m$. Using the triple-product identity (1.7.1) with $\boldsymbol{A} = \boldsymbol{r}$, $\boldsymbol{B} = \boldsymbol{\theta}$, $\boldsymbol{C} = \boldsymbol{r}$,

$$\boldsymbol{r} \times (\boldsymbol{\theta} \times \boldsymbol{r}) = \boldsymbol{\theta}\,(\boldsymbol{r} \cdot \boldsymbol{r}) - \boldsymbol{r}\,(\boldsymbol{r} \cdot \boldsymbol{\theta}) = \boldsymbol{\theta} r^2 - \boldsymbol{r}(\boldsymbol{r} \cdot \boldsymbol{\theta}). \tag{4.3.1}$$

For a spherically symmetric body the second term averages, over the body, to $\tfrac{1}{3}\boldsymbol{\theta} r^2$: indeed $\int x_i x_j \,\mathrm{d}m = \tfrac{1}{3}\delta_{ij}\int r^2 \mathrm{d}m$ by symmetry (the off-diagonal integrals vanish because the integrand is odd, and the three diagonal ones are equal and sum to $\int r^2 \mathrm{d}m$). Hence

$$\boldsymbol{L} = \int \Big(\boldsymbol{\theta} r^2 - \tfrac{1}{3}\boldsymbol{\theta} r^2\Big)\mathrm{d}m = \tfrac{2}{3}\boldsymbol{\theta}\int r^2 \,\mathrm{d}m \equiv I_p \boldsymbol{\theta}, \qquad I_p = \tfrac{2}{3}\int r^2 \mathrm{d}m. \tag{4.3.2}$$

Evaluation for a uniform sphere of radius a and density ρ_s. Write $\mathrm{d}m = \rho_s \,\mathrm{d}V = \rho_s\, 4\pi r^2 \,\mathrm{d}r$ for a spherical shell of radius r and thickness $\mathrm{d}r$ (its volume is surface area $\times$ thickness). Then

$$\int r^2 \,\mathrm{d}m = \int_0^a r^2\, \rho_s 4\pi r^2 \,\mathrm{d}r = 4\pi\rho_s \int_0^a r^4 \,\mathrm{d}r = 4\pi\rho_s \frac{a^5}{5}, \tag{4.3.3}$$

$$I_p = \frac{2}{3}\cdot\frac{4\pi\rho_s a^5}{5} = \frac{8\pi\rho_s a^5}{15}. \tag{4.3.4}$$

It is convenient to express this through the particle mass $m_p = \rho_s \cdot \tfrac{4}{3}\pi a^3$:

$$I_p = \frac{8\pi\rho_s a^5}{15} = \frac{2}{5}\left(\frac{4}{3}\pi a^3 \rho_s\right) a^2 = \frac{2}{5} m_p a^2, \tag{4.3.5}$$

where the middle step is verified by expanding: $\tfrac{2}{5}\cdot\tfrac{4}{3}\pi a^3\rho_s a^2 = \tfrac{8}{15}\pi\rho_s a^5$. ✓

Finally, in terms of the *diameter* $d = 2a$, which is the variable used in Eq. (SM-1):

$$I_p = \frac{8\pi\rho_s}{15}\left(\frac{d}{2}\right)^5 = \frac{8\pi\rho_s}{15}\cdot\frac{d^5}{32} = \frac{\pi\rho_s d^5}{60}. \tag{4.3.6}$$

Keep (4.3.6): it is one of the two ingredients of Eq. (SM-1).

4.4 Magnetic moment, its torque and its energy

Definition 4.4.1 (Magnetic moment)**.** A small current loop, or a uniformly magnetised grain, is characterised by a vector $\boldsymbol{m}$ — its *magnetic moment* — of magnitude equal to the product of the magnetisation and the volume, directed along the magnetisation. For a grain of spontaneous magnetisation M_s and volume V,

$$m = M_s V \tag{4.4.1}$$

(this is the relation used explicitly after Eq. (SM-29)). Units: $\mathrm{G\,cm^3} = \mathrm{erg\,G^{-1}}$.

Result 4.4.1 (Torque and energy of a dipole in a field)**.**

$$\boldsymbol{T} = \boldsymbol{m} \times \boldsymbol{H}, \qquad U = -\boldsymbol{m}\cdot\boldsymbol{H} = -mH\cos\theta. \tag{4.4.2}$$

Derivation of the torque. Model the moment as a planar rectangular current loop of sides b and c carrying current $\mathcal{I}$, so that $m = \mathcal{I}bc/c_{\text{light}}$ in Gaussian units; only the final structure matters here. Place it in a uniform field $\boldsymbol{H}$ with its normal $\hat{\boldsymbol{n}}$ at angle θ to $\boldsymbol{H}$. Each element of the loop feels the Ampère force $\mathrm{d}\boldsymbol{F} \propto \mathrm{d}\boldsymbol{l} \times \boldsymbol{H}$. For the two sides parallel to the axis about which the loop can tilt, the forces are equal and opposite (so the net force vanishes for a uniform field) but they do not act along the same line: they are separated by the distance $b\sin\theta$ projected perpendicular to $\boldsymbol{H}$. Their moment is therefore

$$T = (\text{force}) \times (\text{lever arm}) = \big(\mathcal{I}cH/c_{\text{light}}\big) \times \big(b\sin\theta\big) = mH\sin\theta, \tag{4.4.3}$$

directed so as to rotate $\boldsymbol{m}$ towards $\boldsymbol{H}$. Both the magnitude $mH\sin\theta$ and that direction are precisely what the cross product $\boldsymbol{m} \times \boldsymbol{H}$ encodes. ■

Derivation of the energy. Work must be done against the torque to increase θ:

$$U(\theta) - U(\tfrac{\pi}{2}) = \int_{\pi/2}^{\theta} T\,\mathrm{d}\theta' = \int_{\pi/2}^{\theta} (-mH\sin\theta')\,\mathrm{d}\theta' = mH\cos\theta'\Big|_{\pi/2}^{\theta} = mH\cos\theta, \tag{4.4.4}$$

Careful with the sign: the torque *restores* θ towards zero, so the torque component along increasing θ is $-mH\sin\theta$. Choosing the zero of energy at $\theta = \pi/2$ (moment perpendicular to field) we would obtain $U = +mH\cos\theta$, which has its minimum at $\theta = \pi$ — wrong. The error is that the work done *by* the field is $-\Delta U$; performing the bookkeeping correctly,

$$\mathrm{d}U = -T_{\text{field}}\,\mathrm{d}\theta = +mH\sin\theta\,\mathrm{d}\theta \implies U = -mH\cos\theta + \text{const}, \tag{4.4.5}$$

and with the constant chosen as zero, $U = -\boldsymbol{m}\cdot\boldsymbol{H}$, minimal when $\boldsymbol{m} \parallel \boldsymbol{H}$, as it must be. ■

The combination that will govern the statistical mechanics is the ratio of this energy scale to the thermal energy,

$$\xi \equiv \frac{mH}{kT}, \tag{4.4.6}$$

the *Langevin parameter*: dimensionless, positive, proportional to the field. It is the single most important variable of both papers.

4.5 Magnetisation of a suspension

If there are n particles per unit volume, each with moment m along a unit vector $\boldsymbol{e}$ that fluctuates by thermal agitation, the magnetisation — the magnetic moment per unit volume — is

$$\boldsymbol{M} = n\,m\,\langle \boldsymbol{e} \rangle, \tag{4.5.1}$$

where $\langle\cdot\rangle$ denotes an average over the particles contained in a physically small element of volume: small compared with the scale over which the flow varies, but containing enough particles for the average to be smooth. This "physically small element" idea, used verbatim by both papers, is the bridge between the single particle and the continuum. Note the immediate consequence

$$M \le nm = nM_sV = \varphi M_s, \tag{4.5.2}$$

the saturation magnetisation of the suspension, with $\varphi = nV$ the volume fraction of solid.

Chapter 5

The Equations of Fluid Motion

5.1 Conservation of mass: the continuity equation

Consider a fixed region $\mathcal{V}$ in space, bounded by $\mathcal{S}$. The mass inside is $\int_{\mathcal{V}} \rho \, \mathrm{d}V$; mass leaves only by being carried across the boundary, at the rate $\oint_{\mathcal{S}} \rho \boldsymbol{v} \cdot \boldsymbol{n} \, \mathrm{d}S$. Mass is neither created nor destroyed, so

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathcal{V}} \rho \, \mathrm{d}V = - \oint_{\mathcal{S}} \rho \boldsymbol{v} \cdot \boldsymbol{n} \, \mathrm{d}S. \tag{5.1.1}$$

Because $\mathcal{V}$ is fixed, the time derivative may be taken inside as a partial derivative; and the right-hand side may be converted by the divergence theorem (2.3.4) with $\boldsymbol{A} = \rho \boldsymbol{v}$:

$$\int_{\mathcal{V}} \frac{\partial \rho}{\partial t} \, \mathrm{d}V = - \int_{\mathcal{V}} \nabla \cdot (\rho \boldsymbol{v}) \, \mathrm{d}V \implies \int_{\mathcal{V}} \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{v}) \right] \mathrm{d}V = 0. \tag{5.1.2}$$

Since $\mathcal{V}$ was arbitrary — the equality must hold for every region, including arbitrarily small ones — the integrand must vanish pointwise:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{v}) = 0. \tag{5.1.3}$$

(The step "integral zero for all volumes ⇒ integrand zero" is valid for continuous integrands: if the integrand were positive at a point it would be positive in a small ball around it, and the integral over that ball would be positive.)

Assumption 5.1.1 (Incompressibility)**.** The carrier liquid is water, whose density changes by less than 10^{-4} under the pressures of the experiment. We therefore set $\rho = \text{const}$, whereupon (5.1.3) reduces to

$$\nabla \cdot \boldsymbol{v} = 0. \tag{5.1.4}$$

This is the first of the three relations quoted as Eq. (SM-9).

5.2 Conservation of momentum: the Cauchy equation

Apply Newton's second law to the fluid contained, at a given instant, in a volume $\mathcal{V}$. Its momentum is $\int \rho \boldsymbol{v} \, \mathrm{d}V$, and the forces on it are of two kinds:

- *body forces* $\boldsymbol{f}$ per unit volume, acting on the interior (gravity; in a ferrofluid, the magnetic force $(\boldsymbol{M} \cdot \nabla)\boldsymbol{H}$);

- *surface forces* exerted by the surrounding fluid across $\mathcal{S}$.

The surface force per unit area (the *traction*) on a surface element with normal $\boldsymbol{n}$ is a vector $\boldsymbol{t}(\boldsymbol{n})$ that depends on the orientation $\boldsymbol{n}$. Cauchy's insight is that this dependence must be linear:

$$t_i = \sigma_{ij} n_j, \tag{5.2.1}$$

which *defines* the stress tensor σ_{ij} (units $\mathrm{dyn\,cm^{-2}}$). The linearity follows from applying Newton's law to a shrinking tetrahedron: the surface terms scale as $(\text{length})^2$ while mass$\times$acceleration scales as $(\text{length})^3$, so in the limit the surface forces must balance among themselves, which forces the linear relation (5.2.1).

Newton's second law for $\mathcal{V}$ then reads

$$\int_{\mathcal{V}} \rho \,\frac{\mathrm{D}v_i}{\mathrm{D}t}\,\mathrm{d}V = \int_{\mathcal{V}} f_i \,\mathrm{d}V + \oint_{\mathcal{S}} \sigma_{ij} n_j \,\mathrm{d}S. \tag{5.2.2}$$

The material derivative appears on the left because we are following the fluid (Sec. 3.2). Convert the surface integral by the divergence theorem, applied to the vector field $A_j = \sigma_{ij}$ for each fixed i:

$$\oint_{\mathcal{S}} \sigma_{ij} n_j \,\mathrm{d}S = \int_{\mathcal{V}} \partial_j \sigma_{ij} \,\mathrm{d}V. \tag{5.2.3}$$

Since $\mathcal{V}$ is arbitrary, the integrands match:

$$\boxed{\rho \,\frac{\mathrm{D}v_i}{\mathrm{D}t} = f_i + \partial_j \sigma_{ij}} \tag{5.2.4}$$

This is the Cauchy equation of motion — Newton's second law for a continuum. Everything specific to a particular material now hides in σ_{ij}.

5.3 The Newtonian constitutive law

Assumption 5.3.1 (Newtonian fluid)**.** The stress depends linearly on the *rate* of deformation, isotropically, and not at all on the deformation itself (a fluid has no memory of its shape):

$$\sigma_{ij} = -p\,\delta_{ij} + 2\eta\, e_{ij} \qquad \text{with} \qquad e_{ij} = \tfrac{1}{2}(\partial_j v_i + \partial_i v_j), \tag{5.3.1}$$

for an incompressible fluid. Here p is the pressure and η the dynamic viscosity.

Three remarks that will matter enormously later:

1. The term $-p\delta_{ij}$ survives when the fluid is at rest: a fluid at rest supports only a normal, isotropic stress. The minus sign is because a positive pressure *pushes* on a surface, i.e. acts opposite to its outward normal.

2. Only the symmetric part e_{ij} appears. The antisymmetric part a_{ij} is a rigid rotation (Sec. 3.4), which cannot produce stress in an ordinary fluid, since rigidly rotating a fluid does not deform it.

3. **This last statement fails in a ferrofluid.** If a magnetic torque makes the particles spin at a rate $\boldsymbol{\Theta} \neq \boldsymbol{\Omega}$ different from the fluid, angular momentum is exchanged between the internal (spin) and external (orbital) degrees of freedom, and the stress tensor acquires an *antisymmetric* part. That extra piece is exactly the term $\frac{1}{2}\nabla \times (\boldsymbol{M} \times \boldsymbol{H})$ which appears in Eqs. (SM-12) and (B-3), and it is the origin of the entire negative-viscosity effect. We flag it here and derive it in Volume II.

5.4 The Navier–Stokes equation

Insert (5.3.1) into (5.2.4):

$$\partial_j \sigma_{ij} = \partial_j\big(-p\delta_{ij}\big) + \partial_j\big(2\eta e_{ij}\big) \tag{5.4.1}$$
$$= -\partial_i p + \eta\,\partial_j\big(\partial_j v_i + \partial_i v_j\big) \qquad (\eta \text{ constant};\ \delta_{ij}\partial_j = \partial_i) \tag{5.4.2}$$
$$= -\partial_i p + \eta\,\partial_j\partial_j v_i + \eta\,\partial_i\big(\partial_j v_j\big) \qquad (\text{exchanging } \partial_j\partial_i = \partial_i\partial_j) \tag{5.4.3}$$
$$= -\partial_i p + \eta\,\nabla^2 v_i, \tag{5.4.4}$$

the last term having vanished by incompressibility (5.1.4). Hence

$$\boxed{\rho\left[\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{v}\cdot\nabla)\boldsymbol{v}\right] = -\nabla p + \eta\nabla^2\boldsymbol{v} + \boldsymbol{f}} \tag{5.4.5}$$

— the incompressible Navier–Stokes equation. Comparing with the fluid-motion equations actually used in the papers, Eq. (SM-6) and Eq. (B-3),

$$\rho\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + (\boldsymbol{M}\cdot\nabla)\boldsymbol{H} + \eta\Delta\boldsymbol{v} + \frac{I}{2\tau_s}\nabla\times(\boldsymbol{\Theta}-\boldsymbol{\Omega}), \tag{SM-6}$$

we can now read the structure completely: $\rho\,\mathrm{d}\boldsymbol{v}/\mathrm{d}t$ is the inertia term of (5.4.5) with the material derivative (3.2.3); $-\nabla p$ and $\eta\Delta\boldsymbol{v}$ (their Δ is our ∇^2) are the ordinary Newtonian terms; $(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}$ is the body force $\boldsymbol{f}$ exerted by a *non-uniform* magnetic field on the magnetised medium; and the final term is the new, ferrofluid-specific antisymmetric stress announced in remark 3 above. Every symbol of (SM-6) has now been accounted for except the last term, whose derivation is the business of Volume II.

5.5 The Reynolds number and the Stokes limit

Compare the inertial term $\rho(\boldsymbol{v}\cdot\nabla)\boldsymbol{v}$, of order $\rho U^2/\ell$, with the viscous term $\eta\nabla^2\boldsymbol{v}$, of order $\eta U/\ell^2$, where U and ℓ are the characteristic velocity and length:

$$\mathrm{Re} \equiv \frac{\rho U^2/\ell}{\eta U/\ell^2} = \frac{\rho U\ell}{\eta}. \tag{5.5.1}$$

For a colloidal particle, $\ell \sim d = 10^{-6}$ cm, and the relevant velocity is that of its surface when it rotates at the fluid vorticity, $U \sim \Omega d/2 \sim 60\times 5\times 10^{-7} \approx 3\times 10^{-5}$ cm s^{-1}. With $\rho \approx 1$ g cm^{-3} and $\eta = 10^{-2}$ P,

$$\mathrm{Re} \sim \frac{1\times 3\times 10^{-5}\times 10^{-6}}{10^{-2}} = 3\times 10^{-9} \lll 1. \tag{5.5.2}$$

Inertia is utterly negligible at the scale of a particle. The Navier–Stokes equation then reduces to the *Stokes equations*

$$0 = -\nabla p + \eta\nabla^2\boldsymbol{v}, \qquad \nabla\cdot\boldsymbol{v} = 0, \tag{5.5.3}$$

which are linear — a decisive simplification, and the basis of the next chapter.

Chapter 6

Stokes Flow Around a Rotating Sphere

6.1 The problem and why we must solve it

A magnetic grain forced to rotate at angular velocity $\boldsymbol{\theta}$ inside a liquid at rest drags the liquid with it and feels a retarding torque. The relation between the two,

$$\boldsymbol{T}_{\text{visc}} = -\zeta_r \boldsymbol{\theta}, \tag{6.1.1}$$

defines the *rotational friction coefficient* ζ_r, and *every* quantitative statement in both papers depends on its value: it fixes τ_s in Eq. (SM-1), τ_B in Eq. (SM-5), the constant $6\eta\varphi$ in Eqs. (SM-19), (B-1) and (SM-37), and the numerical prefactor $\frac{3}{2}\eta\varphi$ in Eq. (SM-2). So we derive it in full.

6.2 Setting up

Let a rigid sphere of radius a, centred at the origin, rotate with constant angular velocity $\boldsymbol{\theta}$ in an unbounded liquid that is at rest at infinity. We must solve the Stokes equations (5.5.3) subject to

$$\boldsymbol{v} = \boldsymbol{\theta} \times \boldsymbol{r} \qquad \text{at } r = a \qquad \text{(no-slip on the sphere)}, \tag{6.2.1}$$

$$\boldsymbol{v} \to \boldsymbol{0} \qquad \text{as } r \to \infty \qquad \text{(fluid at rest far away)}. \tag{6.2.2}$$

The no-slip condition (6.2.1) is the experimental fact that a viscous liquid does not slide over a solid surface: the fluid in contact with the sphere moves with the sphere's surface, whose velocity is $\boldsymbol{\theta} \times \boldsymbol{r}$ by Sec. 3.4, Step 1.

6.3 The solution

Result 6.3.1.

$$\boldsymbol{v}(\boldsymbol{r}) = \frac{a^3}{r^3}\,\boldsymbol{\theta} \times \boldsymbol{r}, \qquad p = const \tag{6.3.1}$$

solves the problem.

Step 1: it satisfies the boundary conditions. At $r = a$ the prefactor $a^3/r^3 = 1$, so $\boldsymbol{v} = \boldsymbol{\theta} \times \boldsymbol{r}$: (6.2.1) holds. As $r \to \infty$, $|\boldsymbol{v}| \sim \theta a^3/r^2 \to 0$: (6.2.2) holds.

Step 2: it is incompressible. In components, $v_i = a^3\varepsilon_{ijk}\theta_j x_k r^{-3}$, hence

$$\partial_i v_i = a^3\varepsilon_{ijk}\theta_j\,\partial_i\big(x_k r^{-3}\big) = a^3\varepsilon_{ijk}\theta_j\left[\delta_{ik}r^{-3} + x_k\,\partial_i r^{-3}\right]. \tag{6.3.2}$$

We shall need $\partial_i r^{-3}$ repeatedly. Since $r = (x_l x_l)^{1/2}$,

$$\partial_i r = \frac{1}{2}(x_l x_l)^{-1/2}\cdot 2x_i = \frac{x_i}{r}, \qquad \Longrightarrow \qquad \partial_i r^{-3} = -3r^{-4}\,\partial_i r = -3\frac{x_i}{r^5}. \tag{6.3.3}$$

Substituting into (6.3.2),

$$\partial_i v_i = a^3\theta_j\left[\varepsilon_{ijk}\delta_{ik}r^{-3} - 3\varepsilon_{ijk}\frac{x_k x_i}{r^5}\right] = 0, \tag{6.3.4}$$

because $\varepsilon_{ijk}\delta_{ik} = \varepsilon_{iji} = 0$ (two equal indices) and $\varepsilon_{ijk}x_k x_i = 0$ (antisymmetric contracted with symmetric).

Step 3: it satisfies the Stokes equation with constant pressure. Observe first that

$$\frac{\boldsymbol{\theta}\times\boldsymbol{r}}{r^3} = -\boldsymbol{\theta}\times\nabla\left(\frac{1}{r}\right) = \nabla\times\left(\frac{\boldsymbol{\theta}}{r}\right), \tag{6.3.5}$$

which is verified as follows. From (6.3.3) with the exponent -1, $\partial_i r^{-1} = -x_i/r^3$, i.e. $\nabla(1/r) = -\boldsymbol{r}/r^3$; therefore $-\boldsymbol{\theta}\times\nabla(1/r) = \boldsymbol{\theta}\times\boldsymbol{r}/r^3$. ✓ For the second equality, with $\boldsymbol{\theta}$ a constant vector,

$$\left[\nabla\times\frac{\boldsymbol{\theta}}{r}\right]_i = \varepsilon_{ijk}\partial_j\big(\theta_k r^{-1}\big) = \varepsilon_{ijk}\theta_k\left(-\frac{x_j}{r^3}\right) = \frac{1}{r^3}\varepsilon_{ikj}\theta_k x_j = \frac{(\boldsymbol{\theta}\times\boldsymbol{r})_i}{r^3}, \tag{6.3.6}$$

where one interchange of the last two indices of ε supplied the compensating minus sign. ✓

Now take the Laplacian of (6.3.1) written in the form $\boldsymbol{v} = a^3\nabla\times(\boldsymbol{\theta}/r)$. Since ∇^2 commutes with $\nabla\times$ (both are built from partial derivatives, which commute),

$$\nabla^2\boldsymbol{v} = a^3\,\nabla\times\left(\boldsymbol{\theta}\,\nabla^2\frac{1}{r}\right) = \mathbf{0}, \tag{6.3.7}$$

because $1/r$ is a harmonic function away from the origin:

$$\nabla^2\frac{1}{r} = \partial_i\partial_i r^{-1} = \partial_i\Big(-\frac{x_i}{r^3}\Big) = -\left[\frac{\delta_{ii}}{r^3} + x_i\left(-3\frac{x_i}{r^5}\right)\right] = -\left[\frac{3}{r^3} - \frac{3r^2}{r^5}\right] = 0 \quad (r>0). \tag{6.3.8}$$

So $\eta\nabla^2\boldsymbol{v} = \mathbf{0}$, and (5.5.3) is satisfied by $\nabla p = \mathbf{0}$, i.e. constant pressure, which we set to zero (a constant pressure exerts no net force or torque on a closed surface). ■

By the uniqueness theorem for Stokes flow (two solutions with the same boundary data would differ by a flow with zero boundary velocity and hence zero dissipation, therefore zero), (6.3.1) is *the* solution.

6.4 The traction on the sphere

We need $t_i = \sigma_{ij}n_j$ at $r = a$, with $\boldsymbol{n} = \boldsymbol{r}/a$ the outward unit normal and $\sigma_{ij} = -p\delta_{ij} + \eta(\partial_j v_i + \partial_i v_j)$ (from (5.3.1), writing $2e_{ij}$ out). With $p = 0$,

$$t_i = \eta\big(\partial_j v_i\big)n_j + \eta\big(\partial_i v_j\big)n_j. \tag{6.4.1}$$

We evaluate the two pieces separately. First,

$$\partial_j v_i = a^3 \varepsilon_{ikl} \theta_k \, \partial_j \big(x_l r^{-3} \big) = a^3 \varepsilon_{ikl} \theta_k \left[\frac{\delta_{lj}}{r^3} - 3 \frac{x_l x_j}{r^5} \right]. \tag{6.4.2}$$

Contract with $n_j = x_j/a$ and set $r = a$ (so $x_j = a n_j$, $x_l x_j n_j = x_l a$):

$$\begin{aligned}
\big(\partial_j v_i\big) n_j &= a^3 \varepsilon_{ikl} \theta_k \left[\frac{n_l}{a^3} - 3 \frac{(a n_l)(a n_j) n_j}{a^5} \right] && (6.4.3)\\
&= a^3 \varepsilon_{ikl} \theta_k \left[\frac{n_l}{a^3} - 3 \frac{a^2 n_l}{a^5} \right] && \text{using } n_j n_j = 1 \quad (6.4.4)\\
&= a^3 \varepsilon_{ikl} \theta_k \, \frac{n_l - 3 n_l}{a^3} && (6.4.5)\\
&= -2 \, \varepsilon_{ikl} \theta_k n_l = -2 (\boldsymbol{\theta} \times \boldsymbol{n})_i. && (6.4.6)
\end{aligned}$$

Second,

$$\partial_i v_j = a^3 \varepsilon_{jkl} \theta_k \left[\frac{\delta_{li}}{r^3} - 3 \frac{x_l x_i}{r^5} \right], \tag{6.4.7}$$

so, again at $r = a$ with $x_i = a n_i$,

$$\begin{aligned}
\big(\partial_i v_j\big) n_j &= a^3 \varepsilon_{jkl} \theta_k n_j \left[\frac{\delta_{li}}{a^3} - 3 \frac{(a n_l)(a n_i)}{a^5} \right] && (6.4.8)\\
&= \varepsilon_{jkl} \theta_k n_j \delta_{li} - 3 \, \varepsilon_{jkl} \theta_k n_j n_l n_i && (6.4.9)\\
&= \varepsilon_{jki} \theta_k n_j - 0 && \text{since } \varepsilon_{jkl} n_j n_l = 0 \quad (6.4.10)\\
&= -\varepsilon_{ikj} \theta_k n_j && \text{one interchange } j \leftrightarrow i \text{ in } \varepsilon_{jki} = -\varepsilon_{ikj} \quad (6.4.11)\\
&= -(\boldsymbol{\theta} \times \boldsymbol{n})_i. && (6.4.12)
\end{aligned}$$

Adding (6.4.6) and (6.4.12) in (6.4.1),

$$\boldsymbol{t} = \eta \big[-2 (\boldsymbol{\theta} \times \boldsymbol{n}) - (\boldsymbol{\theta} \times \boldsymbol{n}) \big] = -3\eta \, (\boldsymbol{\theta} \times \boldsymbol{n}). \tag{6.4.13}$$

The traction is everywhere *opposite* to the local surface velocity $a(\boldsymbol{\theta} \times \boldsymbol{n})$: the liquid resists, as it must.

6.5 The torque

The torque about the centre is $\boldsymbol{T} = \oint \boldsymbol{r} \times \boldsymbol{t} \, \mathrm{d}S$ with $\boldsymbol{r} = a\boldsymbol{n}$ on the surface:

$$\begin{aligned}
T_i &= \oint a \, \varepsilon_{ijk} n_j t_k \, \mathrm{d}S = -3\eta a \oint \varepsilon_{ijk} n_j (\boldsymbol{\theta} \times \boldsymbol{n})_k \, \mathrm{d}S && (6.5.1)\\
&= -3\eta a \oint \big[\boldsymbol{n} \times (\boldsymbol{\theta} \times \boldsymbol{n}) \big]_i \, \mathrm{d}S && (6.5.2)\\
&= -3\eta a \oint \Big[\theta_i (\boldsymbol{n} \cdot \boldsymbol{n}) - n_i (\boldsymbol{n} \cdot \boldsymbol{\theta}) \Big] \mathrm{d}S && \text{by (1.7.1)} \quad (6.5.3)\\
&= -3\eta a \left[\theta_i \oint \mathrm{d}S - \theta_j \oint n_i n_j \, \mathrm{d}S \right] && \text{using } \boldsymbol{n} \cdot \boldsymbol{n} = 1. \quad (6.5.4)
\end{aligned}$$

The first integral is the sphere's area, $4\pi a^2$. For the second we use the symmetry result

$$\oint n_i n_j \, \mathrm{d}S = \frac{4\pi a^2}{3} \delta_{ij}, \tag{6.5.5}$$

which is proved exactly as in Sec. 4.3: by symmetry the integral must be proportional to δ_{ij} (there is no preferred direction on a sphere), say $C\delta_{ij}$; taking the trace, $\oint n_i n_i \,\mathrm{d}S = \oint 1 \,\mathrm{d}S = 4\pi a^2$ on the left and $3C$ on the right, so $C = 4\pi a^2/3$. Substituting into (6.5.4),

$$T_i = -3\eta a \left[4\pi a^2 \theta_i - \frac{4\pi a^2}{3} \theta_j \delta_{ij} \right] = -3\eta a \left[4\pi a^2 \theta_i - \frac{4\pi a^2}{3} \theta_i \right] \tag{6.5.6}$$

$$= -3\eta a \cdot 4\pi a^2 \theta_i \left(1 - \frac{1}{3} \right) = -3\eta a \cdot 4\pi a^2 \theta_i \cdot \frac{2}{3} = -8\pi\eta a^3 \theta_i. \tag{6.5.7}$$

Therefore

$$\boxed{\boldsymbol{T}_{\text{visc}} = -8\pi\eta a^3 \,\boldsymbol{\theta} \;\equiv -\zeta_r \boldsymbol{\theta}, \qquad \zeta_r = 8\pi\eta a^3} \tag{6.5.8}$$

6.6 Two equivalent forms of ζ_r used by the papers

Express ζ_r through the particle *volume* $V = \frac{4}{3}\pi a^3$:

$$\zeta_r = 8\pi\eta a^3 = 6\eta \cdot \frac{4}{3}\pi a^3 = 6\eta V. \tag{6.6.1}$$

(Check the arithmetic: $6 \times \frac{4}{3}\pi = 8\pi$. ✓) And in terms of the diameter $d = 2a$,

$$\zeta_r = 8\pi\eta \frac{d^3}{8} = \pi\eta d^3. \tag{6.6.2}$$

Both forms appear in the papers, usually multiplied by the number density n so as to refer to unit volume of suspension:

$$n\zeta_r = 6\eta \, nV = 6\eta\varphi. \tag{6.6.3}$$

This is the origin of the constant $6\eta\varphi$ in the torque-balance equation (B-1) of Bacri *et al.*,

$$6\eta\varphi\big(\boldsymbol{\omega}_p - \boldsymbol{\Omega}\big) = \boldsymbol{M} \times \boldsymbol{H}, \tag{B-1}$$

and of the identity $I/\tau_s = 6\eta\varphi$ quoted between Eqs. (SM-18) and (SM-19). We shall verify that identity explicitly in the next chapter.

Remark 6.6.1 (Why $8\pi\eta a^3$ and not $6\pi\eta a$). The familiar Stokes drag $\boldsymbol{F} = -6\pi\eta a \boldsymbol{u}$ refers to *translation* and has units of (viscosity)×(length); rotational friction relates a torque to an angular velocity and therefore has units (viscosity)×(length)^3. The two coefficients are different numbers because the flow patterns are different — the rotating sphere's flow (6.3.1) decays as $1/r^2$, far faster than the $1/r$ decay around a translating sphere.

Part III

The Papers, Equation by Equation

Chapter 7

Equation (SM-1): The Inertial Relaxation Time of Particle Rotation

The first equation of Shliomis & Morozov is

$$\tau_s = \frac{\rho_s d^2}{60\,\eta}. \tag{SM-1}$$

We now derive it in the eight-part format.

7.1 Context

The paper's argument begins by recalling why a stationary magnetic field increases the viscosity of a ferrofluid. The chain of reasoning is:

1. Each element of liquid rotates locally with $\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \boldsymbol{v}$ (our Chapter 3).

2. *Without* a field, the suspended particles rotate with the same angular velocity, $\boldsymbol{\Theta} = \boldsymbol{\Omega}$.

3. That equality is not a postulate: it is the outcome of a very fast relaxation. If a particle's spin $\boldsymbol{\Theta}$ momentarily differs from $\boldsymbol{\Omega}$, the viscous torque drives the difference to zero. Equation (SM-1) is the time constant of that decay.

4. The point of computing τ_s is to show that it is *fantastically* small (10^{-11} s), so that particle *inertia* may be dropped everywhere in the theory. This is what later allows Eq. (SM-8) to be replaced by the algebraic relation (SM-11), and it is the reason the moment of inertia I disappears from the final results.

So (SM-1) is a *licence to neglect something*, and its derivation must be quantitative enough to justify that licence.

7.2 Symbols

Symbol	Meaning	Type	Units
τ_s	relaxation time for the particle's angular velocity to equalise with the fluid's	scalar, positive, constant	s
ρ_s	mass density of the solid magnetic material of a particle (subscript s = solid)	scalar constant	$\mathrm{g\,cm^{-3}}$
d	particle diameter	scalar constant	cm
η	dynamic viscosity of the *carrier liquid*	scalar constant	$\mathrm{P = g\,cm^{-1}s^{-1}}$
$\boldsymbol{\Theta}$	macroscopic angular velocity of the particles, averaged over a physically small volume element	vector field	$\mathrm{s^{-1}}$
$\boldsymbol{\Omega}$	local angular velocity of the fluid, $\frac{1}{2}\nabla \times \boldsymbol{v}$	vector field	$\mathrm{s^{-1}}$
I_p	moment of inertia of *one* particle	scalar constant	$\mathrm{g\,cm^2}$
I	sum of the particles' moments of inertia *per unit volume*, $I = nI_p$	scalar constant	$\mathrm{g\,cm^{-1}}$
ζ_r	rotational friction coefficient of one particle, $8\pi\eta a^3$	scalar constant	erg s

A dimensional check of the target formula, performed before any derivation:

$$\left[\frac{\rho_s d^2}{\eta}\right] = \frac{\mathrm{g\,cm^{-3}}\cdot\mathrm{cm^2}}{\mathrm{g\,cm^{-1}s^{-1}}} = \frac{\mathrm{g\,cm^{-1}}}{\mathrm{g\,cm^{-1}s^{-1}}} = \mathrm{s}.\checkmark \tag{7.2.1}$$

So (SM-1) is at least dimensionally a time.

7.3 Assumptions

1. The particle is a *rigid sphere* of uniform density ρ_s and diameter d. (Real grains are faceted crystallites of ≈ 10 nm; the sphere is an idealisation accurate to the accuracy of the numerical prefactor.)

2. The particle is small enough that the flow around it is a Stokes flow, $\mathrm{Re} \lll 1$ (verified numerically in Sec. 5.5).

3. The suspension is dilute enough that each particle feels only the carrier liquid, not its neighbours. (Strictly, $\varphi = 0.2$ in the experiment is not dilute; the papers acknowledge this by using the *measured* zero-field viscosity as the reference η.)

4. The only torque acting is the viscous one. Brownian and magnetic torques are excluded from this particular calculation because we are asking a different question: *if* the spin differs from the fluid's, how fast does the difference decay under viscous friction alone?

5. The fluid vorticity $\boldsymbol{\Omega}$ is constant over the time τ_s. Since we will find $\tau_s \sim 10^{-11}$ s and hydrodynamic times are $\gtrsim 10^{-3}$ s, this is overwhelmingly satisfied.

6. Particle inertia is retained here — it is the whole point — but nowhere else in the theory.

7.4 Mathematical Background: the linear relaxation equation

We shall meet the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{y}{\tau}, \tag{7.4.1}$$

where $\tau > 0$ is a constant. Solve it by separation of variables — legitimate wherever $y \neq 0$:

$$\frac{\mathrm{d}y}{y} = -\frac{\mathrm{d}t}{\tau}, \tag{7.4.2}$$

$$\int_{y_0}^{y} \frac{\mathrm{d}y'}{y'} = -\int_0^t \frac{\mathrm{d}t'}{\tau}, \tag{7.4.3}$$

$$\ln y - \ln y_0 = -\frac{t}{\tau}, \tag{7.4.4}$$

$$\ln \frac{y}{y_0} = -\frac{t}{\tau}, \tag{7.4.5}$$

$$y(t) = y_0\, e^{-t/\tau}. \tag{7.4.6}$$

The solution decays by the factor $e^{-1} \approx 0.368$ in the time τ; after 5τ less than 1% remains. Hence τ deserves the name *relaxation time*, and "inertia is negligible" means precisely "τ is short compared with every other time in the problem". Equation (7.4.6) applies componentwise to a vector equation $\mathrm{d}\boldsymbol{y}/\mathrm{d}t = -\boldsymbol{y}/\tau$, since the three components decouple.

7.5 Step-by-Step Derivation

Step 1: the equation of motion for the particle's spin. Apply (4.2.2) to one particle, whose angular momentum is $I_p\boldsymbol{\Theta}$ by (4.3.2):

$$I_p \frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = \boldsymbol{T}_{\text{total}}. \tag{7.5.1}$$

Step 2: identify the torque. By assumption 4, the only torque is viscous. But (6.5.8) was derived for a sphere rotating in a liquid *at rest at infinity*. Here the surrounding liquid is itself rotating locally at $\boldsymbol{\Omega}$. Because the Stokes equations are linear, we may superpose: work in a frame rotating with the fluid, in which the far liquid is at rest and the sphere spins at $\boldsymbol{\Theta} - \boldsymbol{\Omega}$. (The frame rotates so slowly — $\Omega \lesssim 60\ \mathrm{s}^{-1}$ — that fictitious forces are irrelevant at the particle scale.) Hence

$$\boldsymbol{T}_{\text{visc}} = -\zeta_r\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big). \tag{7.5.2}$$

The sign says: if the particle spins faster than the fluid, the torque slows it down; if slower, the torque speeds it up.

Step 3: assemble the equation of motion.

$$I_p \frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = -\zeta_r\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big). \tag{7.5.3}$$

This is precisely the structure of Eq. (SM-8) of the paper,

$$I \frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = \boldsymbol{M} \times \boldsymbol{H} - \frac{I}{\tau_s}\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big), \tag{SM-8}$$

with the magnetic torque switched off and with everything referred to one particle rather than to unit volume. Comparing the two shows already that the paper's τ_s must equal I/ζ in the corresponding units; we now prove it.

Step 4: change variables to the slip. Define the *slip* (the quantity whose decay we are studying)

$$\boldsymbol{y} \equiv \boldsymbol{\Theta} - \boldsymbol{\Omega}. \tag{7.5.4}$$

Because $\boldsymbol{\Omega}$ is constant over the times of interest (assumption 5), $\mathrm{d}\boldsymbol{y}/\mathrm{d}t = \mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t$, and (7.5.3) becomes

$$I_p \frac{\mathrm{d}\boldsymbol{y}}{\mathrm{d}t} = -\zeta_r \boldsymbol{y} \qquad \Longleftrightarrow \qquad \frac{\mathrm{d}\boldsymbol{y}}{\mathrm{d}t} = -\frac{\boldsymbol{y}}{I_p/\zeta_r}. \tag{7.5.5}$$

Step 5: read off the relaxation time. Comparing (7.5.5) with the standard form (7.4.1),

$$\boldsymbol{y}(t) = \boldsymbol{y}(0)\, e^{-t/\tau_s}, \qquad \boxed{\tau_s = \frac{I_p}{\zeta_r}} \tag{7.5.6}$$

Step 6: insert the two ingredients. From (4.3.6), $I_p = \pi\rho_s d^5/60$; from (6.6.2), $\zeta_r = \pi\eta d^3$. Therefore

$$\tau_s = \frac{\pi\rho_s d^5/60}{\pi\eta d^3} \tag{7.5.7}$$

$$= \frac{\pi\rho_s d^5}{60} \cdot \frac{1}{\pi\eta d^3} \qquad \text{(division = multiplication by the reciprocal)} \tag{7.5.8}$$

$$= \frac{\rho_s d^5}{60\,\eta\, d^3} \qquad (\pi \text{ cancels}) \tag{7.5.9}$$

$$= \frac{\rho_s d^2}{60\,\eta}. \qquad (d^5/d^3 = d^2) \tag{7.5.10}$$

That is

$$\tau_s = \frac{\rho_s d^2}{60\eta}, \tag{SM-1}$$

which is Eq. (SM-1) exactly, prefactor included. ■

Step 7: the per-unit-volume version, and the identity $I/\tau_s = 6\eta\varphi$. The paper writes I for the total moment of inertia *per unit volume*, $I = nI_p$. Dividing numerator and denominator of (7.5.6) by the same factor n leaves the ratio unchanged, so

$$\tau_s = \frac{I_p}{\zeta_r} = \frac{nI_p}{n\zeta_r} = \frac{I}{n\zeta_r}, \tag{7.5.11}$$

whence, using (6.6.3),

$$\boxed{\frac{I}{\tau_s} = n\zeta_r = 6\eta\varphi} \tag{7.5.12}$$

This is exactly the identity $I/\tau_s = 6\eta\varphi$ that Shliomis & Morozov invoke without comment between their Eqs. (SM-18) and (SM-19), and it is what converts the last term of (SM-8) into the friction term of (B-1). It also shows why the combination I/τ_s, rather than I or τ_s separately, is what survives in the final theory: it contains no inertia at all, only viscosity and concentration.

7.6 Consistency Checks

Numerical value. With the paper's numbers $d = 10^{-6}$ cm (= 10 nm), $\rho_s = 6$ g cm^{-3}, $\eta = 10^{-2}$ g cm^{-1}s^{-1}:

$$\tau_s = \frac{6 \times (10^{-6})^2}{60 \times 10^{-2}} = \frac{6 \times 10^{-12}}{0.6} = 10^{-11} \text{ s}. \tag{7.6.1}$$

This reproduces the paper's quoted $\tau_s = 10^{-11}$ s exactly. ✓

Scaling. $\tau_s \propto d^2$: bigger particles carry more inertia relative to friction (inertia $\propto d^5$, friction $\propto d^3$). Doubling the diameter quadruples τ_s — still utterly negligible for any colloid.

Comparison with the other times of the problem.

Time	Value in the experiments	Ratio to τ_s
τ_s (inertia of particle spin)	10^{-11} s	1
τ_B (Brownian reorientation)	1.6×10^{-3} s (Bacri *et al.*)	$\sim 10^8$
$1/\omega$ (field period/2π, at 1 kHz)	1.6×10^{-4} s	$\sim 10^7$
$1/\Omega$ (hydrodynamic)	$\gtrsim 1.7 \times 10^{-2}$ s	$\sim 10^9$

Particle inertia is between seven and nine orders of magnitude faster than everything else. Whatever the particles are doing, they are always at instantaneous mechanical equilibrium.

7.7 Physical Interpretation

Equation (SM-1) says: *the memory of a magnetic nanoparticle's own spin lasts* 10^{-11} *seconds.* Three readings of the same fact:

- **As a competition of two effects.** $\tau_s = I_p/\zeta_r$ is (inertia)/(friction). Inertia tries to keep the particle spinning as it was; friction insists that it spin with the fluid. Friction wins by eight orders of magnitude.

- **As a licence.** Because τ_s is so short, the left-hand side of the spin equation (SM-8), $I\,\mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t \sim I\Theta/\tau_{\text{obs}}$, is smaller than the friction term $I\Theta/\tau_s$ by the factor $\tau_s/\tau_{\text{obs}} \sim 10^{-8}$. This is precisely the estimate the paper makes when it writes "$I\,\mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t \ll I\boldsymbol{\Theta}/\tau_s$, so Eq. (8) has the form (11)". Dropping the inertial term converts a differential equation for $\boldsymbol{\Theta}$ into an *algebraic* one, and this is the single most important simplification of the whole theory.

- **As a statement about what is *not* the cause of the effect.** One might imagine that the negative-viscosity effect involves storing energy in the rotational inertia of the grains, like a flywheel. Equation (SM-1) rules that out absolutely: a 10^{-11} s flywheel stores nothing. The energy that ultimately spins up the flow is stored in the *orientational* degrees of freedom of the magnetic moments and released through the magnetic torque, on the much longer time scale τ_B.

What if the term were removed? If instead we *kept* the inertial term as significant, Eq. (SM-8) would remain a differential equation, the particle spin would lag the torque by a phase, and there would be an additional resonance at $\omega \sim 1/\tau_s \sim 10^{11}$ s^{-1} — i.e. at ~ 16 GHz. Nothing in the 1 kHz experiments of Bacri *et al.* can see that; the physics discussed in these papers lives entirely in the regime $\omega\tau_s \lll 1$.

7.8 Connection to the Next Equation

Equation (SM-1) has established the *reference state*: in the absence of a magnetic field the particles spin with the fluid, $\boldsymbol{\Theta} = \boldsymbol{\Omega}$, essentially instantaneously, and therefore contribute no extra dissipation. The paper now asks what happens when a stationary field is switched on. The field grabs the magnetic moments and holds them; a particle whose moment is held cannot follow the fluid's rotation freely; the slip $\boldsymbol{\Omega} - \boldsymbol{\Theta}$ becomes nonzero; and maintaining a nonzero slip costs energy, which appears as an *extra* viscosity. Quantifying that is the content of the next equation,

$$\Delta\eta = \frac{3}{2}\,\eta\varphi\,\frac{\xi - \tanh\xi}{\xi + \tanh\xi}\,\sin^2\beta, \tag{SM-2}$$

Shliomis's 1972 formula for the rotational viscosity in a stationary field.

Deriving (SM-2) requires three ingredients that do not yet exist in this book and which will be constructed, from first principles, at the start of Volume II:

1. **The Boltzmann distribution and the Langevin function.** We must compute $\langle \boldsymbol{e} \rangle$ for dipoles in a field at temperature T, obtaining $L(\xi) = \coth\xi - 1/\xi$ and hence the equilibrium magnetisation $M_0 = nmL(\xi)$ used in Eq. (SM-10).

2. **Rotational Brownian motion.** We must derive the rotational diffusion (Fokker–Planck/Smoluchowski) equation for the orientation $\boldsymbol{e}$, obtain the Einstein relation $D_r = kT/\zeta_r$, and from it the Brownian time $\tau_B = 3\eta V/kT$ of Eq. (SM-5) — which, note, is already within reach: combining $D_r = kT/\zeta_r$ with $\zeta_r = 6\eta V$ from (6.6.1) gives $\tau_B = 1/(2D_r) = 3\eta V/kT$ immediately.

3. **The link between microscopic torque and macroscopic viscosity.** We must show that the averaged magnetic torque density $\overline{\boldsymbol{M} \times \boldsymbol{H}}$ enters the momentum equation as an antisymmetric stress and hence as an addition $\Delta\eta$ to the viscosity — the step that turns Eq. (SM-26) into (SM-27), and which in the papers appears in the compact forms $\overline{\boldsymbol{M} \times \boldsymbol{H}} = -6\eta\varphi(\boldsymbol{\Omega} - \boldsymbol{\Theta})$ and $\Delta\eta = \frac{3}{2}\eta\varphi(\boldsymbol{\Omega} - \boldsymbol{\Theta})/\boldsymbol{\Omega}$.

Volume II will carry out these three constructions in order and then derive (SM-2), (SM-3), (SM-4) and (SM-5) in full.

Chapter 8

Status and Road Map

8.1 What has been established so far

Result	Statement	Where derived
Index calculus	$\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$ and consequences	Secs. 1.6–1.7
Divergence theorem	$\int \nabla \cdot \boldsymbol{A}\,\mathrm{d}V = \oint \boldsymbol{A} \cdot \boldsymbol{n}\,\mathrm{d}S$	Sec. 2.3
Local flow structure	translation + strain + rigid spin, with $\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \boldsymbol{v}$	Sec. 3.4
Vorticity in the experiment	$\boldsymbol{\Omega} \perp \boldsymbol{H}$ in a capillary along the coil axis; $\Omega\tau_B \approx 0.1$	Sec. 3.5
Sphere inertia	$I_p = \pi\rho_s d^5/60 = \frac{2}{5} m_p a^2$	Sec. 4.3
Dipole mechanics	$\boldsymbol{T} = \boldsymbol{m} \times \boldsymbol{H}$, $U = -\boldsymbol{m} \cdot \boldsymbol{H}$, $\xi = mH/kT$	Sec. 4.4
Continuity	$\nabla \cdot \boldsymbol{v} = 0$ [first relation of (SM-9)]	Sec. 5.1
Cauchy / Navier–Stokes	$\rho\,\mathrm{D}\boldsymbol{v}/\mathrm{D}t = -\nabla p + \eta\nabla^2\boldsymbol{v} + \boldsymbol{f}$	Secs. 5.2–5.4
Stokes regime	$\mathrm{Re} \sim 3 \times 10^{-9}$ at the particle scale	Sec. 5.5
Rotational friction	$\zeta_r = 8\pi\eta a^3 = 6\eta V = \pi\eta d^3$; $n\zeta_r = 6\eta\varphi$	Ch. 6
Eq. (SM-1)	$\tau_s = \rho_s d^2/60\eta = 10^{-11}$ s; $I/\tau_s = 6\eta\varphi$	Ch. 7

8.2 The remaining equations, in the order they will be derived

Volume II and its successors will proceed strictly in the papers' own order. For reference, the full list of what remains:

- **Shliomis & Morozov:** (SM-2) rotational viscosity in a stationary field; (SM-3), (SM-4) its weak- and strong-field limits; (SM-5) $\tau_B = 3\eta V/kT$; (SM-6)–(SM-9) the full ferrohydrodynamic system; (SM-10) the Langevin magnetisation; (SM-11)–(SM-13) elimination of the particle spin; (SM-14)–(SM-15) the alternating field and linear susceptibility; (SM-16)–(SM-24) the circular-polarisation decomposition; (SM-25)–(SM-29) the averaged torque and the central result $\Delta\eta = \frac{1}{4}\eta\varphi\xi^2(1-\omega^2\tau_B^2)/(1+\omega^2\tau_B^2)^2$; (SM-30)–(SM-34) the second, direct variant of the solution and the formula $\Delta\eta = \frac{3}{2}\eta\varphi(\Omega-\Theta)/\Omega$; (SM-35)–(SM-43) arbitrary field amplitude, the functions F, G, $\gamma(t)$, and the numerical results.
- **Bacri *et al.*:** (B-1), (B-2) torque balance and $\Delta\eta$; (B-3), (B-4) the hydrodynamic and

magnetodynamic pair; (B-5)–(B-8) the effective field method of Martsenyuk, Raikher and Shliomis; (B-9) the averaged torque and the function g; (B-10) the grouping of the magnetic term with the viscous one.

Every one of these will be derived here with the same standard of completeness: nothing asserted, nothing skipped, every symbol defined, every approximation justified.

End of Volume I. Volume II begins with the Boltzmann distribution for a dipole in a field and proceeds to Eq. (SM-2).